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# Import libraries
import cv2
import numpy as np
import matplotlib.pyplot as plt
from google.colab import files

# Upload image
uploaded = files.upload()

# Get the uploaded image filename
image_path = list(uploaded.keys())[0]

# Read the image
img = cv2.imread(image_path)

# Check if image is loaded successfully
if img is None:
    print("Error: Unable to load image.")
else:
    # Convert image to grayscale
    gray_image = cv2.cvtColor(img, cv2.COLOR_BGR2GRAY)

    # Create a 5x5 rectangular kernel
    kernel = cv2.getStructuringElement(cv2.MORPH_RECT, (5, 5))

    # Apply erosion
    eroded_image = cv2.erode(gray_image, kernel, iterations=1)

    # Save the output image
    cv2.imwrite("eroded_image.jpg", eroded_image)

    # Display original and eroded images
    plt.figure(figsize=(10, 5))

    plt.subplot(1, 2, 1)
    plt.imshow(gray_image, cmap='gray')
    plt.title("Original Grayscale Image")
    plt.axis("off")

    plt.subplot(1, 2, 2)
    plt.imshow(eroded_image, cmap='gray')
    plt.title("Eroded Image")
    plt.axis("off")

    plt.show()

    print("Output image saved as 'eroded image.jpg'")
```

exp 5 cv.webp

exp 5 cv.webp(image/webp) - 234854 bytes, last modified: 22/7/2026 - 100% done

Saving exp 5 cv.webp to exp 5 cv.webp

Original Grayscale Image



Eroded Image



Output image saved as 'eroded_image.jpg'